VIT BHOPAL

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Problem Statement:

Real estate valuation is a somewhat opaque and inconsistent practice. For many people, it can be difficult to discern the fair market value of a piece of property without conducting extensive manual research of existing listings, or by seeking professional appraisal. The problem this project aims to solve is the absence of a readily available, automated solution for instantaneously estimating a piece of property's value based on physical characteristics (square footage, number of rooms), even when a similar instance does not exist.

Why It Matters:

In today's housing market, data-driven decision-making is critical

- **For Buyers: It allows them to make sure they're not paying too much.**
- **For Sellers: It helps them price their home competitively but fairly.**
- **Technical Significance: From a developer's perspective, this is a great starting point to introduce the concept of Supervised Learning to a computer, using historical data to train it to understand relationships between data variables, in this case, how size affects cost.**

Approach & Methodology:

I developed this project using a tiered logic approach to guarantee precision and versatility:

- **Data Structure: I employed the Pandas library to build a structured Data Frame with historical houses data.**

- **Tier 1: Exact Match Logic: The system first conducts a "hard search." If the user provides a square footage and room count that already exists in our records, the system will return the actual historical price. This guarantees 100% precision for data that already exists**

- **Tier 2: Predictive Modeling (Linear Regression): If the system fails to find an exact match, it will then rely on a Scikit-Learn Linear Regression predictive model. This will then determine the line of best fit across all data points to estimate a price for the "gap" in the data.**

- **Tier 3: Data Visualization: To provide context to the user, Matplotlib will then plot a scatter plot of the data. This will then enable the user to understand where their query falls in relation to the broader market trend.**

Key Decisions:

Algorithm Choice: I selected Linear Regression because it is known that square footage and price have a linear relationship (i.e., as square footage increases, so does price). Linear Regression is also computationally simple and easy to interpret.

User Interface: I selected a Command-Line Interface (CLI) for inputs in order to focus on the back-end logic. However, I used a Graphical User Interface (GUI) popup box to display the results obtained using Matplotlib in order to make it easier to digest.

Input Validation: One major decision was to include error checking in my code. The code is set up to not allow non-numeric inputs (e.g., "five" instead of "5") in order to avoid crashing the model in execution.

Challenges Faced:

- **Data Scarcity: With the limited data available, the Linear Regression model might get affected by 'outliers,' i.e., data points that are far from the mean, such as a house that is very small but expensive because of the location. I had to ensure that the data was well-balanced.**

- **Library Integration: At first, it was a bit tricky to match the dimensions of the user's input data with the Scikit-Learn library's requirement for the. predict () function. This was resolved by ensuring proper data reshaping into a 2D array.**

- **Visualization Clarity: To ensure that the user's specific point is clearly identifiable from the rest of the data points plotted, the Matplotlib functions ‘scatter ()' and 'plot ()' had to be fine-tuned.**

Lessons Learned:

- **The Power of Pre-processing: I learned that the way you organize your data in Pandas is just as important as the algorithm you choose.**

- **Hybrid Logic: I also came to understand that AI is not always the solution; "Exact Match" database queries and "Predictions" from AI can be a powerful combination for a reliable application.**

- **Iteration is Key: Experimenting with "different extreme inputs" for the model taught me the power of linear models and has peaked my interest in further investigating Multi-variable Regression for future versions, such as location or year of the house.**

Conclusion:

The House Price Predictor has been able to effectively show how Machine Learning can solve a real-world financial problem. With its ability to bridge the gap between raw data and useful knowledge, it has created a useful tool for basic real estate calculation.